

the Fourier series of the function  $f(x) = x^2, -\pi \leq x \leq \pi$   
 the Fourier Transform of  
 $f(x) = 1 - x^2, |x| \leq 1$   
 $0, |x| \geq 1$   
 Evaluate  $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx$ .  
 Sampling and its types.

DEVI AHILYA UNIVERSITY, INDORE  
 II BE (Civil Engineering)

Class Test I, August 2023

Subject Code: 3VLRC2

Subject Name: Applied Mechanics and Strength of materials

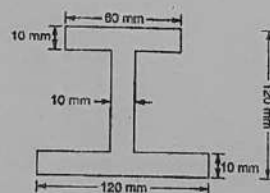
Duration: 70 Min

Maximum Marks: 20

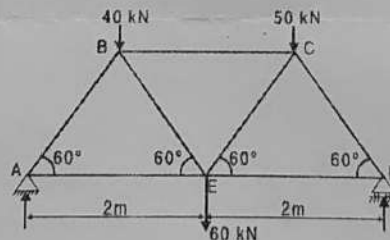
Note: Attempt all questions. All questions carry equal marks. Assume Any Data Missing.

Q1. Derive an expression for moment of inertia of a rectangular lamina of width 'b' and depth 'h'.

Q2. Find the moment of inertia of an I section as shown in figure about X-X axes through the centre of gravity of the section.



Q3. Find the nature and magnitude of forces in different members of truss as shown in fig



Q4. Write a short note. (Any one)

(a) Parallel axis theorem (b) Types of Trusses.